

DIGITAL LOGIC DESIGN

(CE_118)

CHAPTER 3:

Register Transfer Specification & Design (part_1)

Outline

- **Register Transfer (RT) design:**
 - **FSMD model**
 - **RT specification with**
 - **Static-action tables**
 - **ASM charts**
- **Procedure for synthesis from RT specification**
- **Design Optimization through**
 - **Register sharing**
 - **Functional unit sharing**
 - **Bus sharing**
 - **Unit chaining and Multiclocking**
- **Design Pipelining**
 - **Unit pipelining**
 - **Control pipelining**
 - **Datapath pipelining**
- **Scheduling of flowcharts**

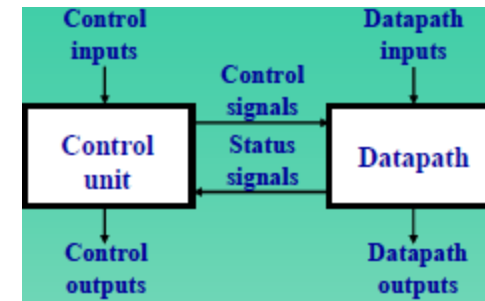
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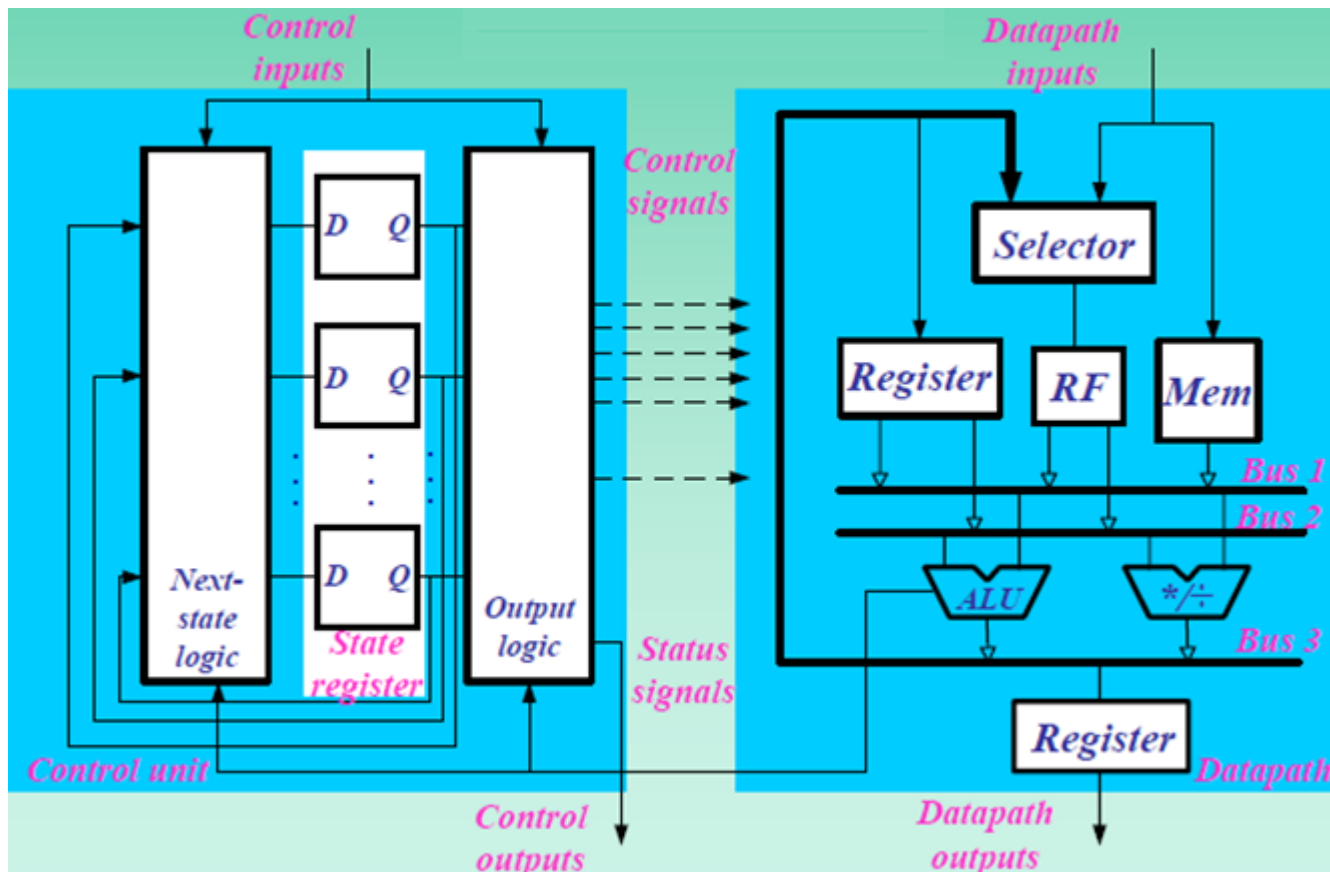
Register-transfer design

- Each standard or custom IC consists of one or more datapaths and control units.
- To synthesize such IC we introduce the model of a FSM with a datapath (FSMD).
- We demonstrate synthesis algorithms for FSMD model, including component selection, resource sharing, pipelining and scheduling.

Design Model

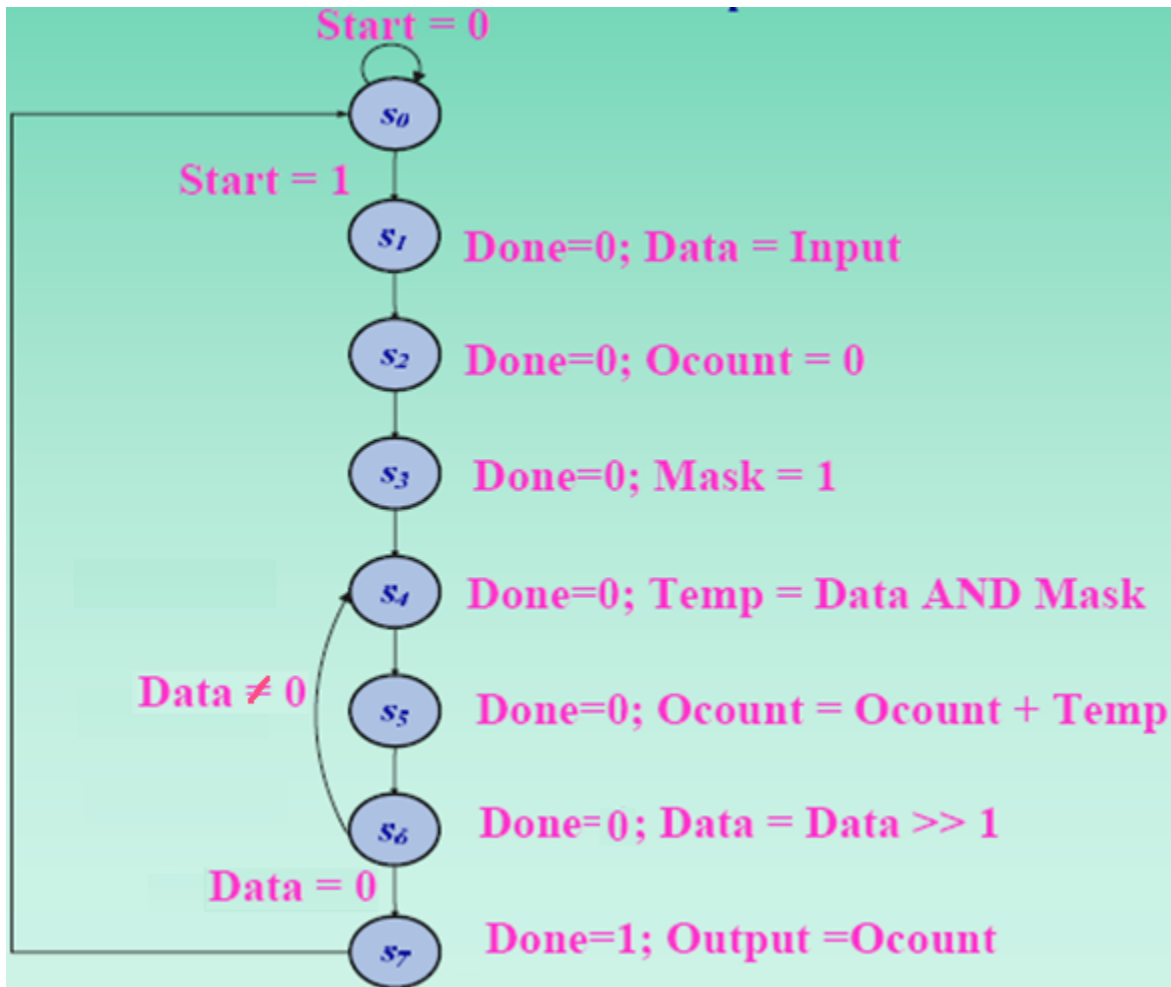
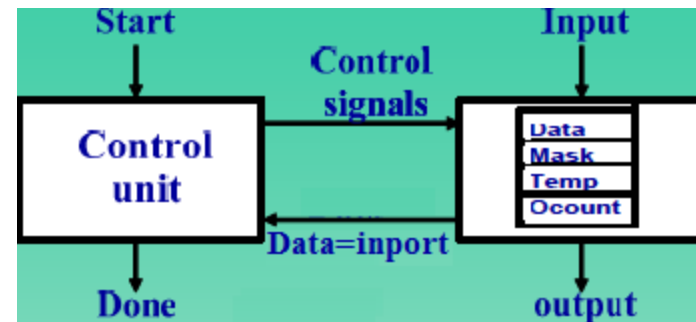


High-level block diagram



Register-transfer-level block diagram

Ones-counter specification



FSDM Definition

- In chapter 1, we defined an FSM as a quintuple $\langle \mathbf{S}, \mathbf{I}, \mathbf{O}, \mathbf{f}, \mathbf{h} \rangle$
 - where $\mathbf{S}$ is a set of states, $\mathbf{I}$ and $\mathbf{O}$ are the sets of input and output symbols:
 $\mathbf{f}: \mathbf{S} \times \mathbf{I} \rightarrow \mathbf{S}$ (next state), and $\mathbf{h}: \mathbf{S} \times \mathbf{I} \rightarrow \mathbf{O}$ (FSM output)
 - More precisely,

□ $\mathbf{I} = A_1 \times A_2 \times \dots \times A_k$ □ $\mathbf{S} = Q_1 \times Q_2 \times \dots \times Q_m$ □ $\mathbf{O} = Y_1 \times Y_2 \times \dots \times Y_n$	- with A_i ($1 \leq i \leq k$) is an input signal - with Q_i ($1 \leq i \leq m$) is the flip-flop output - with Y_i ($1 \leq i \leq n$) is an output signal
--	---
- To define a **FSMD**, we define a set of variables $\mathbf{V} = \mathbf{V}_1 \times \mathbf{V}_2 \times \dots \times \mathbf{V}_q$ which defines the state of the datapath by defining the values of all variables in each state.
 and $\mathbf{I} = \mathbf{I}_C \times \mathbf{I}_D$ $\mathbf{I}_C$: a set of FSM input, $\mathbf{I}_D$: a set of datapath input
 $\mathbf{O} = \mathbf{O}_C \times \mathbf{O}_D$ $\mathbf{O}_C$: a set of FSM output, $\mathbf{O}_D$: a set of datapath output
 - where $\mathbf{I}_C = A_1 \times A_2 \times \dots \times A_k$ as before and $\mathbf{I}_D = B_1 \times B_2 \times \dots \times B_p$,
 - where $\mathbf{O}_C = Y_1 \times Y_2 \times \dots \times Y_n$ as before and $\mathbf{O}_D = Z_1 \times Z_2 \times \dots \times Z_r$

FSDM Definition

- We can simplify function $f : (S \times V) \times I \rightarrow S \times V$ by separating it into two parts: f_C and f_D .
 - The function f_C defines the next state of the control unit

$$f_C : S \times I_C \times STAT \rightarrow S$$
 - The function f_D defines the values of datapath variables in the next state,

$$f_D : S \times V \times I_D \rightarrow V$$

$$f_D := \{f_{Di} : V \times I_D \rightarrow V : \{V_j = e_j \mid V_j \in V, e_j \in Expr(V \times I_D)\}\}$$
- Also, we can do the same thing with: $h : S \times V \times I \rightarrow O$

$$h_C : S \times I_C \times STAT \rightarrow O_C \quad (\text{external control output})$$

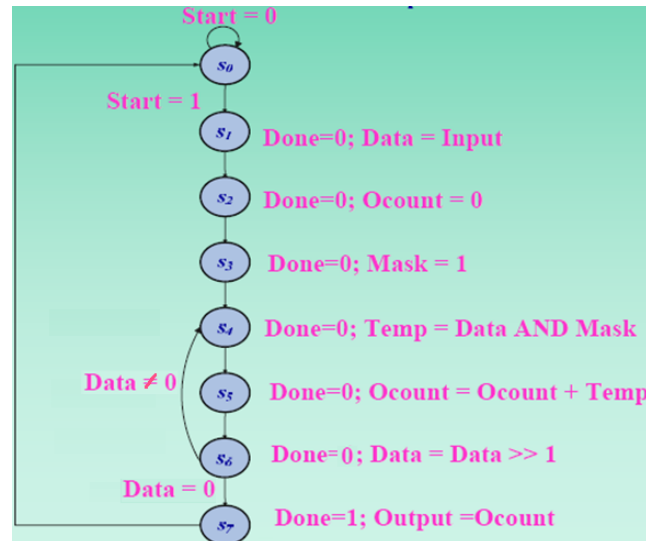
$$h_D : S \times V \times I_D \rightarrow O_D \quad (\text{external datapath output})$$

STAT: set of all status signals --.> relations between variables or experection of variables

FSMD specification of Ones-counter

```

1. Data := Inport
2. Ocount := 0
3. Mask := 1
while Data ≠ 0 repeat
  4. Temp := Data AND Mask
  5. Ocount := Ocount + Temp
  6. Data := Data >> 1
end while
7. Output := Ocount
Basic algorithm for one's counter
  
```



PRESENT STATE	NEXT STATE (<i>Start</i> , <i>Data</i> = 0)				CONTROL OUTPUT	DATAPATH OUTPUT	DATAPATH VARIABLES			
	00	01	10	11			<i>Data</i>	<i>Ocount</i>	<i>Temp</i>	<i>Mask</i>
<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₁	<i>s</i> ₁	0	Z	X	X	X	X
<i>s</i> ₁	<i>s</i> ₂	<i>s</i> ₂	<i>s</i> ₂	<i>s</i> ₂	0	Z	<i>Inport</i>	X	X	X
<i>s</i> ₂	<i>s</i> ₃	<i>s</i> ₃	<i>s</i> ₃	<i>s</i> ₃	0	Z	<i>Data</i>	0	X	X
<i>s</i> ₃	<i>s</i> ₄	<i>s</i> ₄	<i>s</i> ₄	<i>s</i> ₄	0	Z	<i>Data</i>	<i>Ocount</i>	X	1
<i>s</i> ₄	<i>s</i> ₅	<i>s</i> ₅	<i>s</i> ₅	<i>s</i> ₅	0	Z	<i>Data</i>	<i>Ocount</i>	<i>Data</i> AND <i>Mask</i>	<i>Mask</i>
<i>s</i> ₅	<i>s</i> ₆	<i>s</i> ₆	<i>s</i> ₆	<i>s</i> ₆	0	Z	<i>Data</i>	<i>Ocount</i> + <i>Temp</i>	X	<i>Mask</i>
<i>s</i> ₆	<i>s</i> ₄	<i>s</i> ₇	<i>s</i> ₄	<i>s</i> ₇	0	Z	<i>Data</i> >> 1	<i>Ocount</i>	X	<i>Mask</i>
<i>s</i> ₇	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	1	<i>Ocount</i>	<i>Data</i>	<i>Ocount</i>	X	X

State and output table

FSMD specification of Ones-counter (cont.)

```

1. Data := Inport
2. Ocount := 0
3. Mask := 1
while Data ≠ 0 repeat
  4. Temp := Data AND Mask
  5. Ocount := Ocount + Temp
  6. Data := Data >> 1
end while
7. Output := Ocount
Basic algorithm for one's counter

```

PRESENT STATE	NEXT STATE (<i>Start</i> , <i>Data</i> = 0)				CONTROL OUTPUT	DATAPATH OUTPUT	DATAPATH VARIABLES			
	00	01	10	11			<i>Data</i>	<i>Ocount</i>	<i>Temp</i>	<i>Mask</i>
<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₁	<i>s</i> ₁	0	Z	X	X	X	X
<i>s</i> ₁	<i>s</i> ₂	<i>s</i> ₂	<i>s</i> ₂	<i>s</i> ₂	0	Z	<i>Inport</i>	X	X	X
<i>s</i> ₂	<i>s</i> ₃	<i>s</i> ₃	<i>s</i> ₃	<i>s</i> ₃	0	Z	<i>Data</i>	0	X	X
<i>s</i> ₃	<i>s</i> ₄	<i>s</i> ₄	<i>s</i> ₄	<i>s</i> ₄	0	Z	<i>Data</i>	<i>Ocount</i>	X	1
<i>s</i> ₄	<i>s</i> ₅	<i>s</i> ₅	<i>s</i> ₅	<i>s</i> ₅	0	Z	<i>Data</i>	<i>Ocount</i>	<i>Data</i> AND <i>Mask</i>	<i>Mask</i>
<i>s</i> ₅	<i>s</i> ₆	<i>s</i> ₆	<i>s</i> ₆	<i>s</i> ₆	0	Z	<i>Data</i>	<i>Ocount</i> + <i>Temp</i>	X	<i>Mask</i>
<i>s</i> ₆	<i>s</i> ₄	<i>s</i> ₇	<i>s</i> ₄	<i>s</i> ₇	0	Z	<i>Data</i> >> 1	<i>Ocount</i>	X	<i>Mask</i>
<i>s</i> ₇	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	1	<i>Ocount</i>	<i>Data</i>	<i>Ocount</i>	X	X

State and output table

PRESENT STATE	NEXT STATE (<i>Start</i> , <i>Data</i> = 0)				CONTROL OUTPUT	DATAPATH OUTPUT	DATAPATH VARIABLES
	00	01	10	11			
<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₁	<i>s</i> ₁	0	Z	
<i>s</i> ₁	<i>s</i> ₂	<i>s</i> ₂	<i>s</i> ₂	<i>s</i> ₂	0	Z	<i>Data</i> = <i>Inport</i>
<i>s</i> ₂	<i>s</i> ₃	<i>s</i> ₃	<i>s</i> ₃	<i>s</i> ₃	0	Z	<i>Ocount</i> = 0
<i>s</i> ₃	<i>s</i> ₄	<i>s</i> ₄	<i>s</i> ₄	<i>s</i> ₄	0	Z	<i>Mask</i> = 1
<i>s</i> ₄	<i>s</i> ₅	<i>s</i> ₅	<i>s</i> ₅	<i>s</i> ₅	0	Z	<i>Temp</i> = <i>Data</i> AND <i>Mask</i>
<i>s</i> ₅	<i>s</i> ₆	<i>s</i> ₆	<i>s</i> ₆	<i>s</i> ₆	0	Z	<i>Ocount</i> = <i>Ocount</i> + <i>Temp</i>
<i>s</i> ₆	<i>s</i> ₄	<i>s</i> ₇	<i>s</i> ₄	<i>s</i> ₇	0	Z	<i>Data</i> = <i>Data</i> >> 1
<i>s</i> ₇	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	<i>s</i> ₀	1	<i>Ocount</i>	

State and output table with variable assignments

FSMD specification of Ones-counter (cont.)

PRESENT STATE	NEXT STATE (Start, Data = 0)				CONTROL OUTPUT	DATAPATH OUTPUT	DATAPATH VARIABLES
	00	01	10	11	Done	Output	
s_0	s_0	s_0	s_1	s_1	0	Z	
s_1	s_2	s_2	s_2	s_2	0	Z	$Data = Inport$
s_2	s_3	s_3	s_3	s_3	0	Z	$Ocount = 0$
s_3	s_4	s_4	s_4	s_4	0	Z	$Mask = 1$
s_4	s_5	s_5	s_5	s_5	0	Z	$Temp = Data \text{ AND } Mask$
s_5	s_6	s_6	s_6	s_6	0	Z	$Ocount = Ocount + Temp$
s_6	s_4	s_7	s_4	s_7	0	Z	$Data = Data \gg 1$
s_7	s_0	s_0	s_0	s_0	1	Ocount	

State and output table with variable assignments

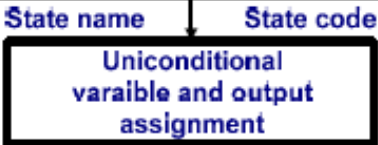
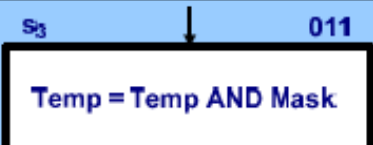
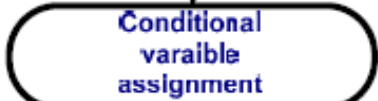
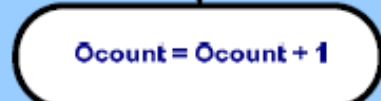
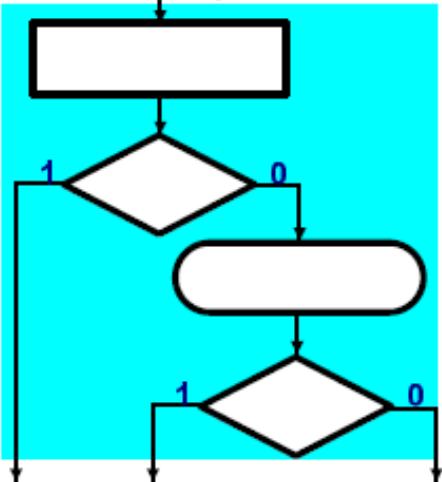
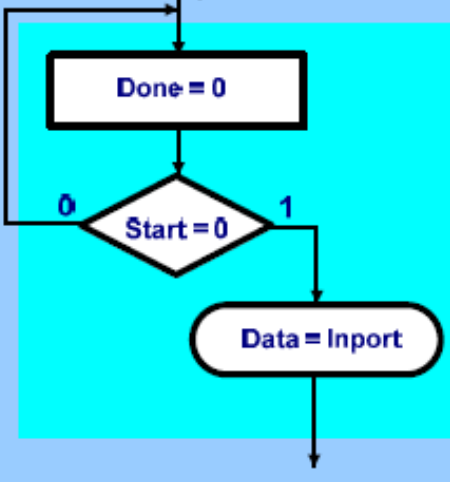
PRESENT STATE	NEXT STATE	CONTROL AND DATAPATH ACTIONS
	CONDITION, STATE	CONDITION, ACTIONS
s_0	$\left[\begin{array}{l} Start = 0, \\ Start = 1, \end{array} \begin{array}{l} s_0 \\ s_1 \end{array} \right]$	$\left[\begin{array}{l} Done = 0 \\ Output = Z \end{array} \right]$
s_1	s_2	$Data = Inport$
s_2	s_3	$Ocount = 0$
s_3	s_4	$Mask = 1$
s_4	s_5	$Temp = Data \text{ AND } Mask$
s_5	s_6	$Ocount = Ocount + Temp$
s_6	$\left[\begin{array}{l} Data \neq 0, \\ Data = 0, \end{array} \begin{array}{l} s_4 \\ s_7 \end{array} \right]$	$Data = Data \gg 1$
s_7	s_0	$\left[\begin{array}{l} Done = 1 \\ Output = Ocount \end{array} \right]$

State-action table

Algorithmic-State-Machine (ASM) chart

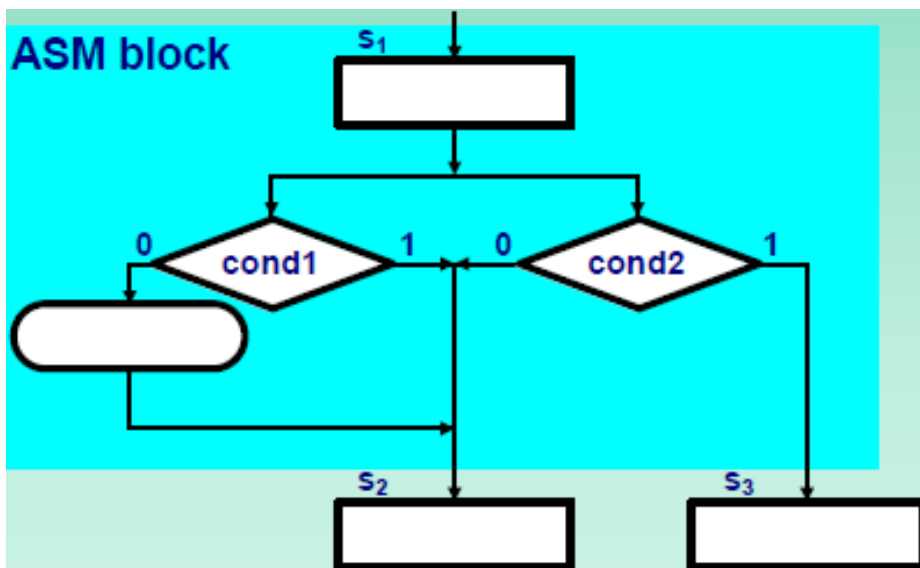
- Graphic representation of FSMD model
- Equivalent to state-action table
- Similar to a flowchart used for program description

ASM Symbols

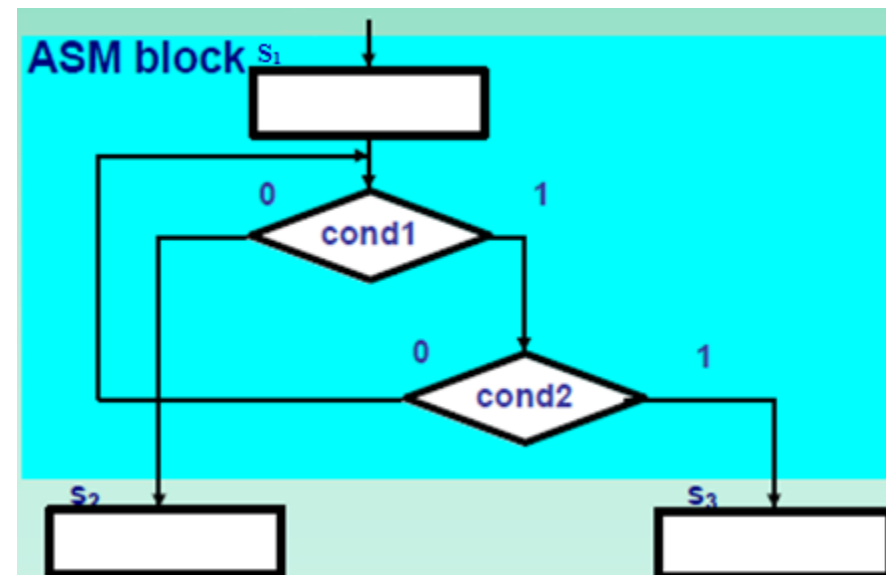
Name	Definition	Example
State box		
Decision Box		
Condition Box		
ASM Block		

ASM rules

- **Rule 1:** The chart must define a unique next state for each state and set of conditions.
- **Rule 2:** Every path defined by the network of condition boxes (**decision boxes**) must lead to another state.

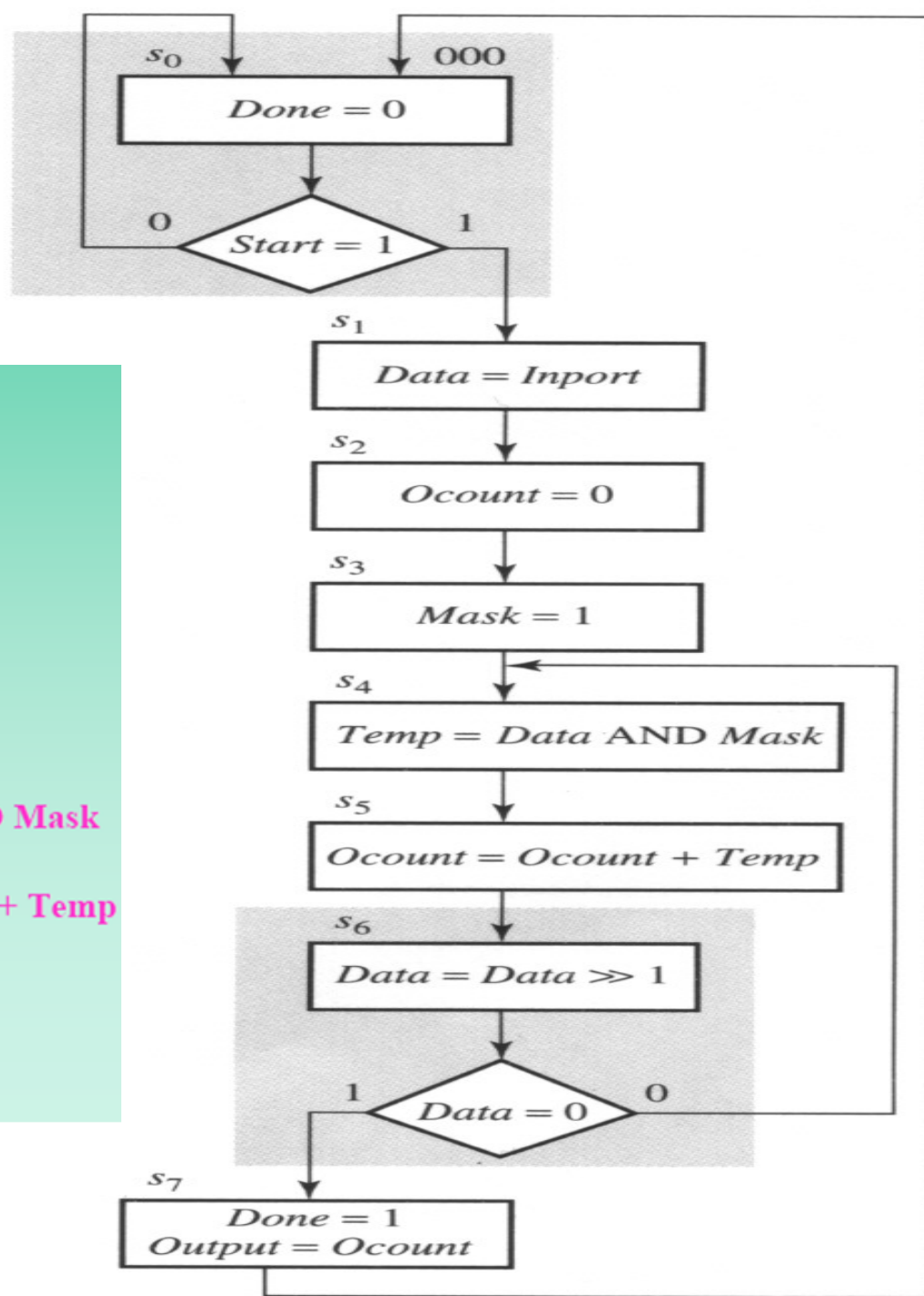


Undefined next state



Undefined exit path

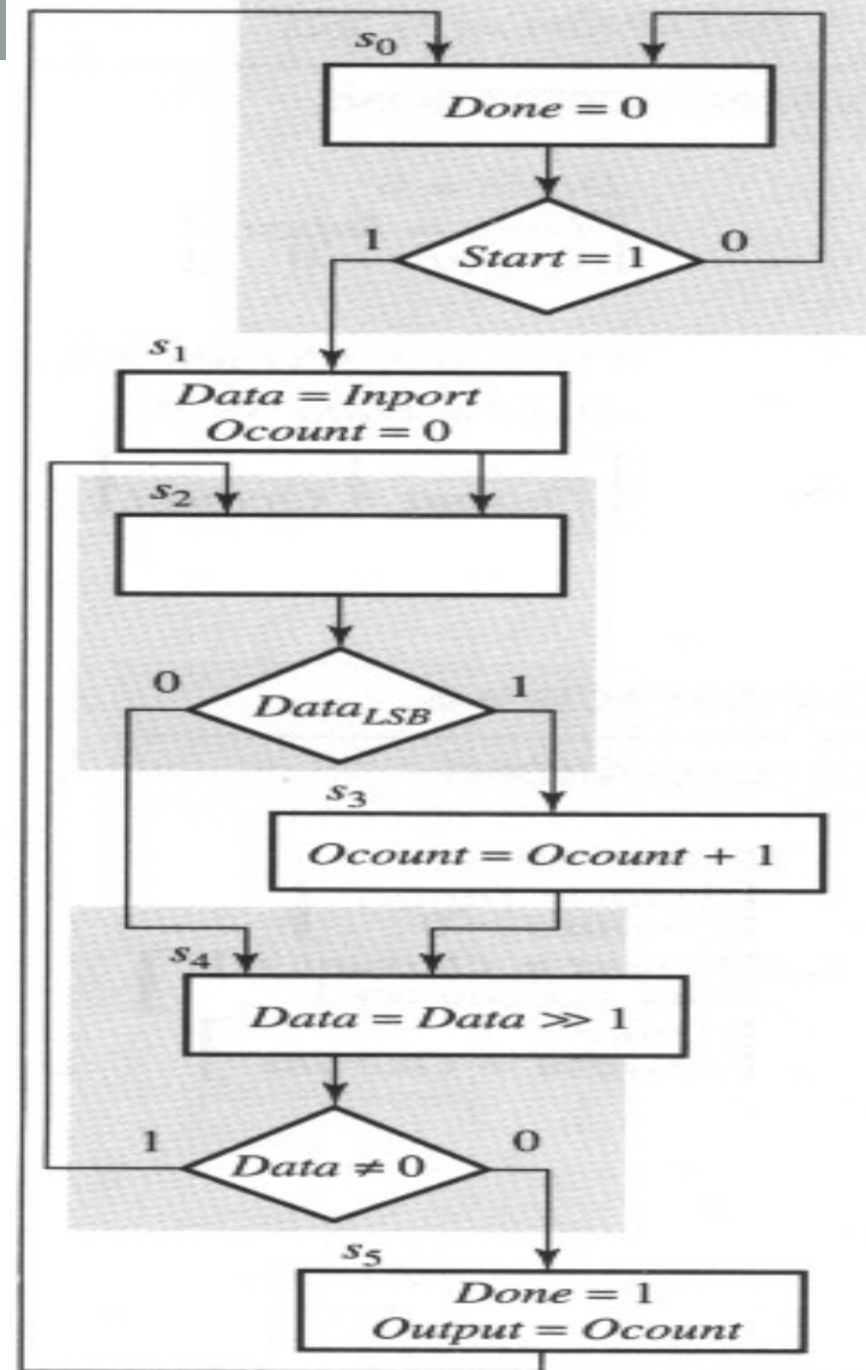
ASM chart for Ones-counter



ASM chart for Ones-counter

Custom Data path

State-based (**Moore**) chart



State-action tables for Ones-counter

PRESENT STATE				NEXT STATE		DATAPATH ACTIONS	
Q_2	Q_1	Q_0	NAME	CONDITION,	STATE	CONDITION,	OPERATIONS
0	0	0	s_0	$[Start = 0,$	$s_0]$		$Done = 0$
0	0	1	s_1	$Start = 1,$	s_1		$[Data = Inport]$
0	1	0	s_2		s_2		$[Ocount = 0]$
0	1	1	s_3	$[Data_{LSB} = 1,$	$s_3]$		
1	0	0	s_4	$Data_{LSB} = 0,$	$s_4]$		$Ocount = Ocount + 1$
1	0	1	s_5	$[Data \neq 0,$	$s_2]$		$Data = Data \gg 1$
				$Data = 0,$	$s_5]$		
					s_0		$[Done = 1]$
							$[Output = Ocount]$

State-based table

$$\begin{aligned}
 D_2 = Q_2(\text{next}) &= s_2 Data'_{LSB} + s_3 + s_4 (Data = 0) \\
 &= Q_1 Q'_0 Data'_{LSB} + Q_1 Q_0 + Q_2 Q'_0 (Data = 0)
 \end{aligned}$$

$$\begin{aligned}
 D_1 = Q_1(\text{next}) &= s_1 + s_2 Data_{LSB} + s_4 (Data \neq 0) \\
 &= Q'_2 Q'_1 Q_0 + Q_1 Q'_0 Data_{LSB} + Q_2 Q'_0 (Data \neq 0)
 \end{aligned}$$

$$\begin{aligned}
 D_0 = Q_0(\text{next}) &= s_0 Start + s_2 Data_{LSB} + s_4 (Data = 0) \\
 &= Q'_2 Q'_1 Q'_0 Start + Q_1 Q'_0 Data_{LSB} + Q_2 Q'_0 (Data = 0)
 \end{aligned}$$

State-action tables for Ones-counter

$$\mathbf{D}_2 = Q_1 \cancel{Q'_\theta} \text{Data}'_{\text{LSB}} + Q_1 Q_0 + Q_2 Q'_0 (\text{Data} = 0)$$

$$\mathbf{D}_1 = \mathbf{Q}'_2 \mathbf{Q}'_1 \mathbf{Q}_0 + \mathbf{Q}_1 \mathbf{Q}'_0 \text{Data}_{\text{LSB}} + \mathbf{Q}_2 \mathbf{Q}'_0 (\text{Data} \neq 0)$$

$$\mathbf{D}_0 = \mathbf{Q}'_2\mathbf{Q}'_1\mathbf{Q}'_0\text{Start} + \mathbf{Q}_1\mathbf{Q}'_0\text{Data}_{\text{LSB}} + \mathbf{Q}_2\mathbf{Q}'_0(\text{Data} = 0)$$

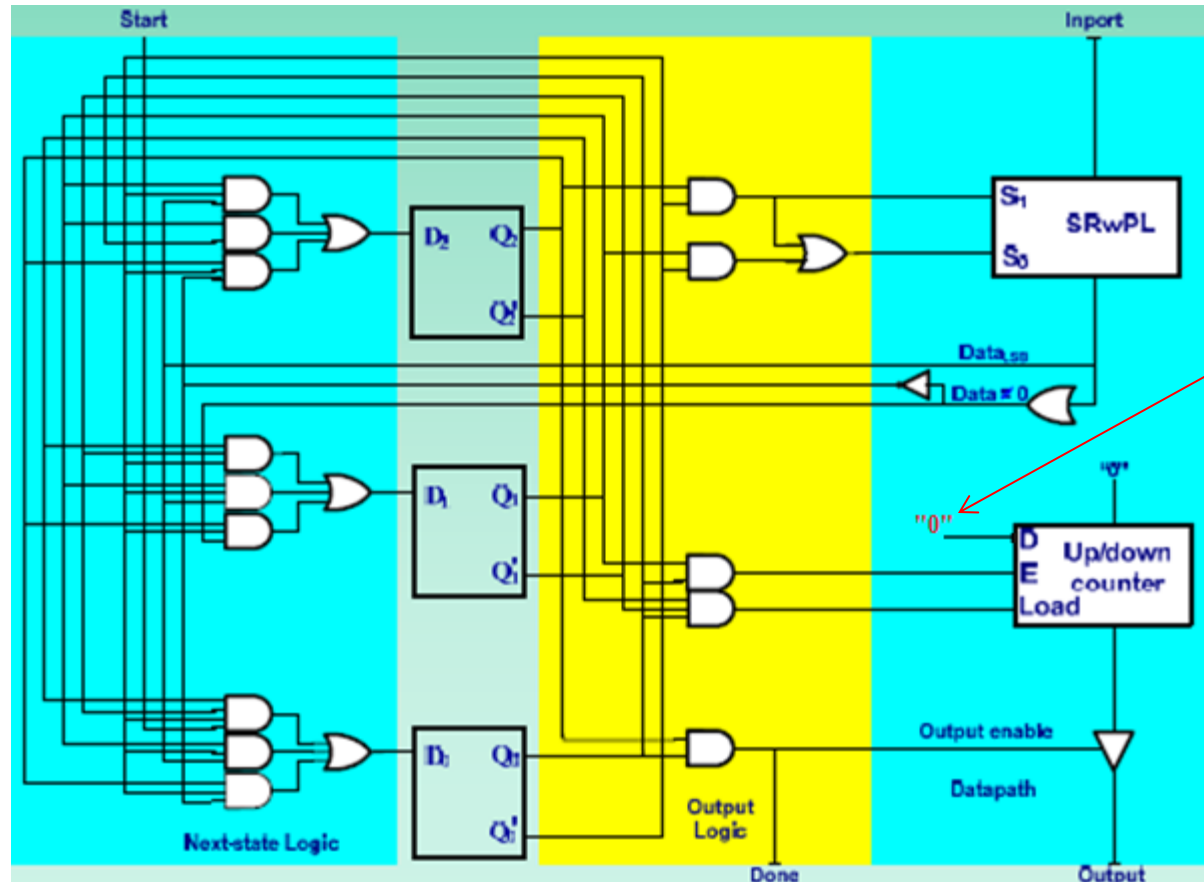
$$S_1 = S_4 = Q_2 Q'_0$$

$$\mathbf{S}_0 = \mathbf{s}_1 + \mathbf{s}_4 = \mathbf{Q}'_2 \mathbf{Q}'_1 \mathbf{Q}'_0 + \mathbf{Q}_2 \mathbf{Q}'_0$$

$$\mathbf{E} = \mathbf{s}_3 = \mathbf{Q}_1 \mathbf{Q}_0$$

Load = $s_1 = Q'_2 Q'_1 Q_0$

Done = Output Enable
= $s_5 = Q_2 Q_0$



s_i : state name

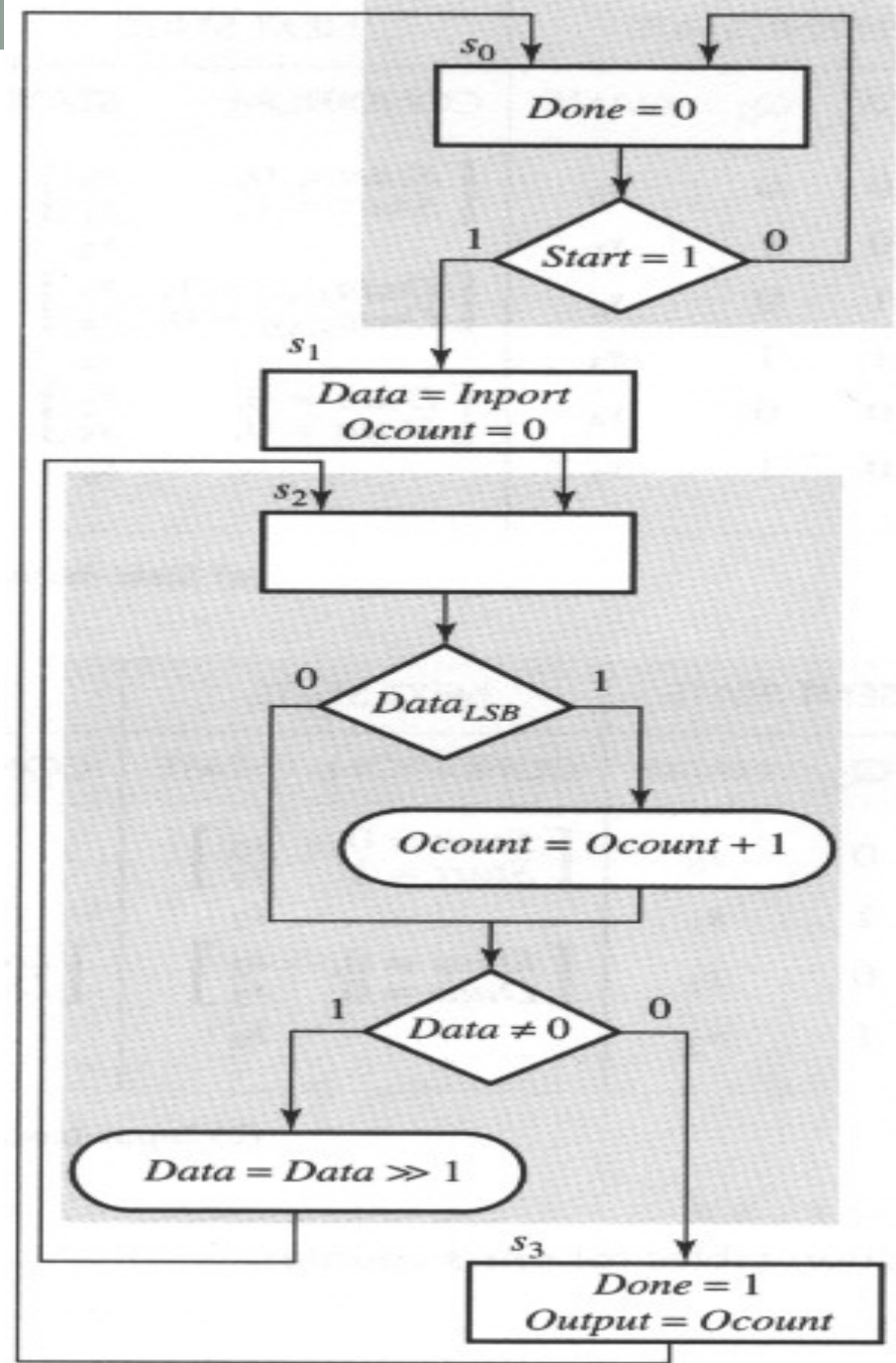
S_i: control signal of SRwPL

State-based version

ASM chart for Ones-counter

Custom Data path (cont.)

Input-based (**Mealy**) chart



State-action tables for Ones-counter

PRESENT STATE			NEXT STATE	DATAPATH ACTIONS	
Q_1	Q_0	NAME	CONDITION, STATE	CONDITION,	OPERATIONS
0	0	s_0	$\left[\begin{array}{l} \text{Start} = 0, \\ \text{Start} = 1, \end{array} \begin{array}{l} s_0 \\ s_1 \end{array} \right]$		$\text{Done} = 0$
0	1	s_1	s_2		$\left[\begin{array}{l} \text{Data} = \text{Inport} \\ \text{Ocount} = 0 \end{array} \right]$
1	0	s_2	$\left[\begin{array}{l} \text{Data} \neq 0, \\ \text{Data} = 0, \end{array} \begin{array}{l} s_2 \\ s_3 \end{array} \right]$	$\left[\begin{array}{l} \text{Data}_{\text{LSB}} = 1, \\ \text{Data} \neq 0, \end{array} \right]$	$\left[\begin{array}{l} \text{Ocount} = \text{Ocount} + 1 \\ \text{Data} = \text{Data} \gg 1 \end{array} \right]$
1	1	s_3	s_0		$\left[\begin{array}{l} \text{Done} = 1 \\ \text{Output} = \text{Ocount} \end{array} \right]$

Input-based table

$$D_1 = Q_1(\text{next}) = s_1 + s_2 = Q'_1 Q_0 + Q_1 Q'_0$$

$$\begin{aligned} D_0 = Q_0(\text{next}) &= s_0 \text{Start} + s_2 (\text{Data} \neq 0)' \\ &= Q'_1 Q'_0 \text{Start} + Q_1 Q'_0 (\text{Data} \neq 0)' \end{aligned}$$

Logic schematics for Ones-counter

$$D_1 = Q'_1 Q_0 + Q_1 Q'_0$$

$$D_0 = Q'_1 Q'_0 \text{Start} + Q_1 Q'_0 (\text{Data} \neq 0)'$$

$$S_1 = s_2 (\text{Data} \neq 0) = Q_1 Q'_0 (\text{Data} \neq 0)$$

$$S_0 = s_1 + s_2 (\text{Data} \neq 0) = Q'_1 Q_0 + Q_1 Q'_0 (\text{Data} \neq 0)$$

$$E = s_2 \text{Data}_{\text{LSB}} = Q_1 Q'_0 \text{Data}_{\text{LSB}}$$

$$\text{Load} = s_1 = Q'_1 Q_0$$

$$\text{Done} = \text{Output enable} = s_3 = Q_1 Q_0$$

Input-based version

